

Rapport mensuel

Étudiant

Superviseur

Posté le

12/06/2026

Tâche(s) principale(s)

During this period, the internship focused on the ongoing migration of the Oversight Cloud platform from Airtable to PostgreSQL/Aurora. The primary objective was to evolve the backend architecture so that the same application logic can operate transparently on either Airtable or PostgreSQL, depending on the selected runtime configuration.

A significant portion of the work was dedicated to the data transport layer and to the implementation of PostgreSQL support for site-related operations. This included database connection management, Airtable-compatible parsing and serialization mechanisms, backend selection through configuration parameters, and improvements to schema conversion and replication tooling. Particular attention was given to the handling of Airtable-specific field types such as links, lookups, scalar values, and JSON structures.

In parallel, testing infrastructure was enhanced to support both storage backends. Smoke and contract tests were adapted to validate identical application behavior against Airtable and PostgreSQL environments, ensuring consistency throughout the migration process.

Another important milestone was the first production release of Oversight Cloud on the new AWS-based infrastructure. Although PostgreSQL integration was not enabled in production, this deployment validated the new cloud architecture and operational workflow while keeping database-related changes isolated from the release.

Additional activities included AWS preproduction environment preparation, DNS and certificate configuration, cleanup of Airtable-related environment variables, and the development of utilities to migrate simulation files from DigitalOcean storage to AWS.

Réalisation(s)

A functional PostgreSQL implementation for site-related operations was successfully developed and validated through automated testing. The backend is now capable of reading and writing data through PostgreSQL while preserving the Airtable-compatible API responses expected by existing application components.

The Airtable-to-Aurora replication process was also significantly improved. New tooling was introduced to validate schema consistency, extract records from Airtable, transform them into PostgreSQL-compatible formats, perform database upserts, and generate detailed migration reports describing successful, updated, and failed operations.

The generated PostgreSQL schema was refined to more accurately reflect Airtable semantics. Several fields previously represented as generic JSON structures were converted into simpler and more explicit column types, improving data clarity and preparing the foundation for a cleaner relational model.

A further achievement was the successful delivery of the first production release on the AWS platform. This deployment confirmed the reliability of the new infrastructure and deployment pipeline while maintaining a low-risk migration strategy.

Operational improvements were also completed for preproduction support, storage migration workflows, and infrastructure configuration required for future migration stages.

Problème(s) rencontré(s) One of the main challenges involved reconciling Airtable's flexible data model with PostgreSQL's structured relational approach. Airtable frequently represents relationships and lookup values as arrays or nested objects, whereas PostgreSQL benefits from more explicit identifiers and normalized data structures. Considerable effort was therefore required to define reliable mappings while preserving application compatibility.

Another difficulty was maintaining a backend-agnostic testing strategy. The test framework had to support both Airtable-based and PostgreSQL-based environments, requiring common fixtures, database initialization procedures, and consistent validation logic across different storage systems.

Preserving compatibility with the existing API contract also proved challenging. Many application components continue to expect Airtable-shaped records, making it necessary to implement conversion layers that transform PostgreSQL rows into formats indistinguishable from the original Airtable responses.

Finally, the production rollout required careful risk management. To minimize operational impact, PostgreSQL integration remained disabled in production while the new AWS deployment model was validated independently.

Prochaine(s) étape(s) The next phase of the project will extend PostgreSQL support beyond sites to the remaining core entities, including organizations, users, simulations, role assignments, documentation records, firmware records, deployment forms, and logs.

Replication and validation processes will be exercised against larger production-like datasets to identify remaining schema inconsistencies, missing field mappings, and edge cases before a broader migration effort.

Work will also continue on reducing Airtable dependencies by progressively replacing Airtable-specific services and workflows with backend-native implementations, particularly in areas such as permissions, authentication, and automation processes.

Finally, the AWS preproduction environment will be used to validate the complete migration workflow, including backend selection, Aurora connectivity, data replication accuracy, simulation file migration, and overall operational readiness before planning a full production cutover.

**Rapport intermédiaire
(un seul par mois)**
